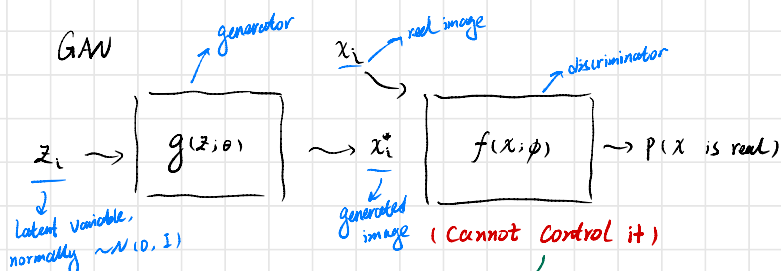


Once GAN is trained, we can sample $z \sim \mathcal{N}(0, I)$ and then pass it to Generator to generate high quality image.

However, we cannot **Control** the image we want to generate.



If we trained on images of animals, and we want to sample image of dogs. Here we can only pass z many times, and get one until it is a dog.

Q:

in standard GAN, why doesn't each feature of z control one part of the image? e.g. z_1 for color, z_2 for background etc.

A:

if this is true, then all possible combinations can happen in a image $\rightarrow$ Violate physical laws generating real images.

because all features are entangled

e.g. if $z_1 \rightarrow$ gender, $z_2 \rightarrow$ beard,

then it is equally likely to generate image $\rightarrow$ male + beard

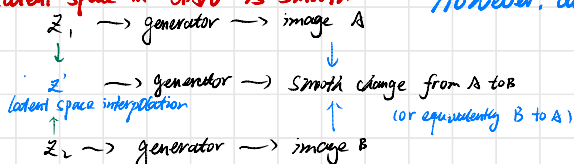
$\rightarrow$ female + beard

$\downarrow$
this is much less realistic

The Latent Variable z encode the information for generating the image.

The generator act as a **decoder**. It generate image from z without disentangle of the features.

The Latent space in GAN is smooth

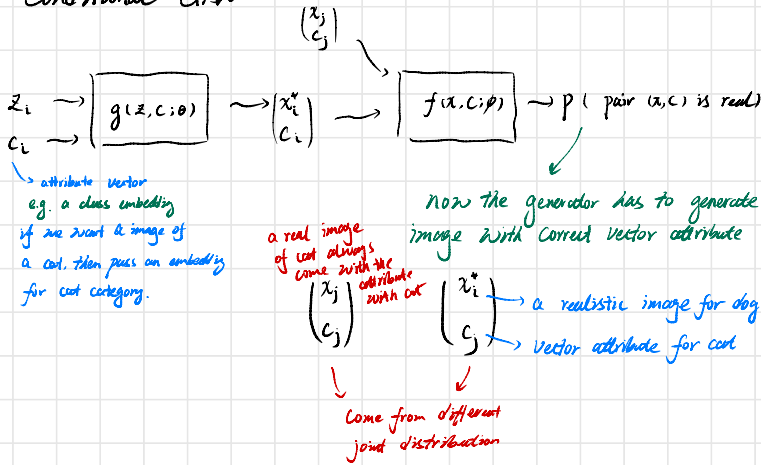


However, all features are entangled.

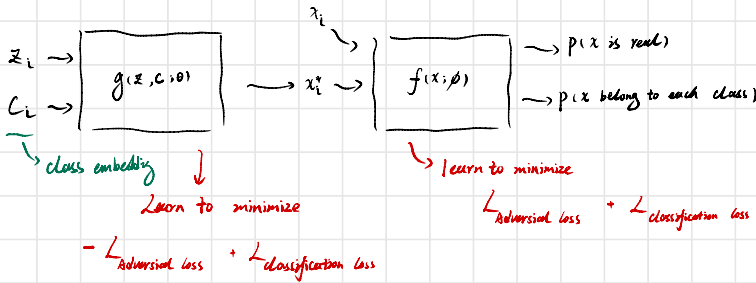
If we change one component of z ,

many part of image A will change (because features are highly entangled)

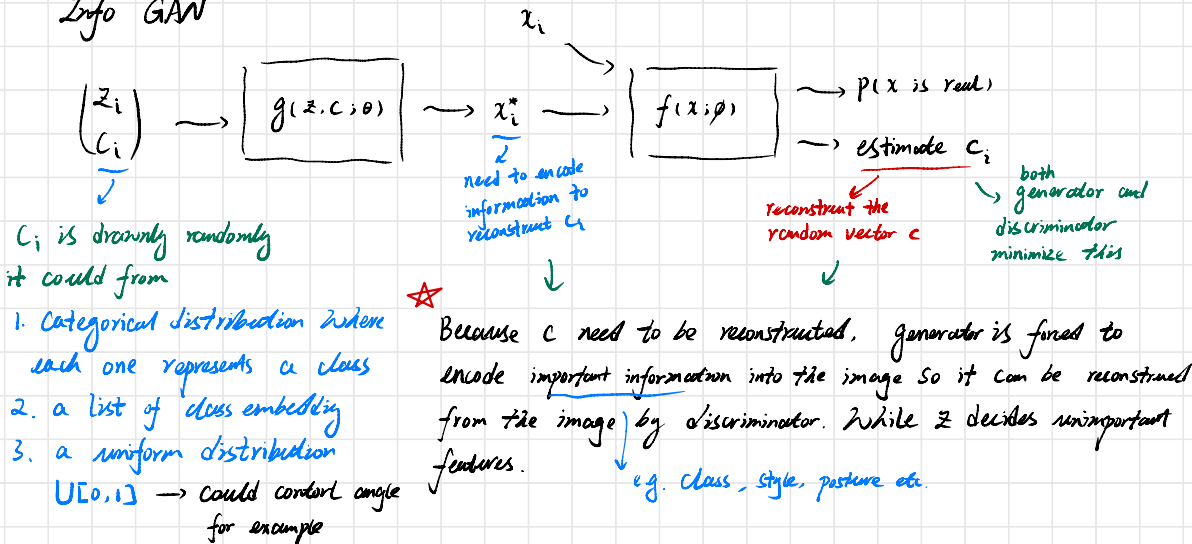
Conditional GAN



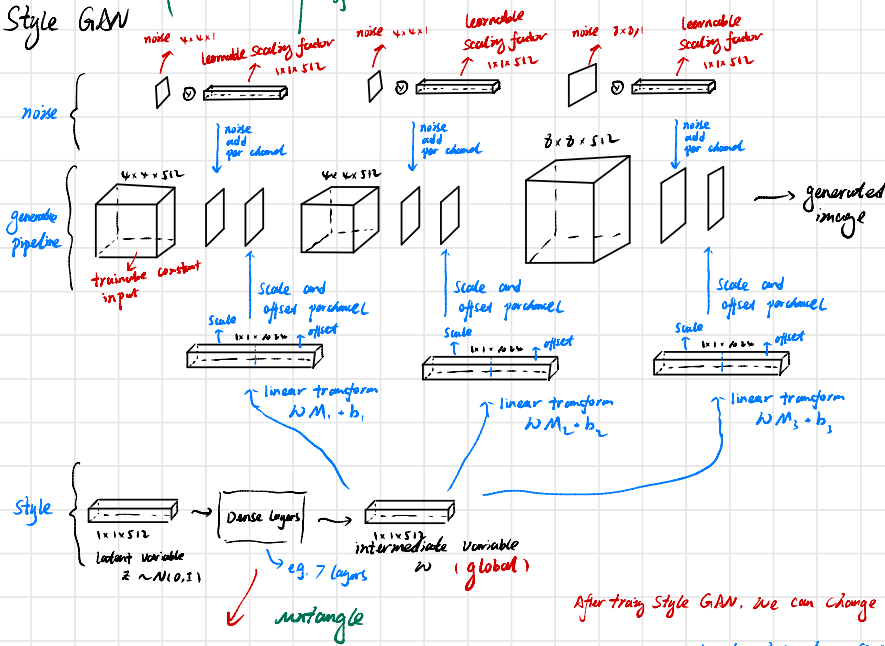
Auxiliary Classifier GAN



Info GAN



Style GAN



After training Style GAN, we can change w to control the generated image.

example: how do we know components in w to control gender?

why do we map $z \rightarrow w$ first?

1: Gaussian noise. It is hard for the model to use one dimension to control one feature.

2: advantage

10: the model learned a new distribution fits w well with real world image.

$z, \dots, z_n \rightarrow \text{generator}$

- images of male $\rightarrow w_1, \dots, w_n$
- images of female $\rightarrow w_1^*, \dots, w_n^*$

gender vector $\tilde{w} = \tilde{w}^*$

$w \rightarrow \text{generator} \rightarrow$ a image of a female

$w + \tilde{w} - \tilde{w}^* \rightarrow \text{generator} \rightarrow$ a image of a male

operation in latent space

Why would Style GAN work?

Implicit assumption: if there are many ways for a neural network to solve a problem, it has the bias to pick the easy one.

intermediate variable w (global) $\rightarrow$ it controls every layer (change of any part of w affect the whole image)

noise for each layer (local) $\rightarrow$ it controls details that can make a image look more realistic, but does not matter that much. e.g. position of hairs. (change of random noise is implemented in pixel-level)